Errata for the TME–EMT Wiki
(Art01, Art06, Art08, Art09)

Compiled during formalization work in
`AlexKontorovich/PrimeNumberTheoremAnd`

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Abstract

This note collects discrepancies found while matching statements on the TME–EMT wiki against the cited source papers, in the course of formalizing those statements in Lean. Items are grouped by wiki page. Only high-confidence mismatches are listed; each entry gives the wiki text, the source, and a proposed correction.

How to read this note

Citations of the form "Art01" refer to Art01.html ("Explicit bounds on primes"). Page numbers for papers are from the versions available at the URLs recorded in the wiki bibliography.

1 Art01 — Explicit bounds on primes

1.1 Deléglise–Nicolas 2019, π_2 upper error constant

Wiki. Art01 quotes, for $x \geq 60173$,

$$\pi_2(x) - \left(\tfrac{x^3}{3\log x} + \tfrac{x^3}{9\log^2 x} + \tfrac{2x^3}{27\log^3 x}\right) \leq \tfrac{11181\,x^3}{648\log^4 x}.$$

Source. Deléglise–Nicolas, *The Landau function and the Riemann hypothesis*, arXiv:1907.07664, Corollary 2.7, display (2.21), uses the constant **1181** (not 11181). The same constant appears in the simplification leading to (2.22), as the coefficient 1181/216.

Proposed correction. Replace 11181 by 1181.

1.2 Platt–Trudgian 2021, ψ versus π

Wiki. Art01 attributes to Platt–Trudgian (2021) the bound

$$\left|\tfrac{\psi(x)-x}{x}\right| \leq 235\,(\log x)^{0.52}\exp\Big(-\sqrt{\tfrac{\log x}{5.573412}}\Big) \qquad (x \geq e^{2000}).$$

Source. Platt–Trudgian, *The Riemann hypothesis is true up to* $3\cdot 10^{12}$, Table 1 at $X = 2000$ and Theorem 1 give a ψ-bound of the shape

$$\left|\tfrac{\psi(x)-x}{x}\right| \leq 411.4\left(\tfrac{\log x}{R}\right)^{1.52}\exp\big(-1.89\sqrt{\log x/R}\big), \qquad R = 5.573412.$$

The constants 235 and exponent 0.52 belong to the companion π-bound (Corollary 2), not to the ψ-bound.

Proposed correction. Quote the ψ-bound from Table 1 / Theorem 1, or explicitly relabel the displayed estimate as a π-bound and cite Corollary 2.

1.3 Johnston–Yang 2023, exponent on $\log x$

Wiki. Art01 quotes

$$\left|\tfrac{\psi(x)-x}{x}\right| \le 9.39\,(\log x)^{1.51}\exp\bigl(-0.8274\sqrt{\log x}\bigr) \qquad (x \ge 2).$$

Source. Johnston–Yang, Theorem 1.1, uses the exponent **1.515**.

Proposed correction. Replace 1.51 by 1.515.

1.4 De Koninck–Letendre 2020, missing k in $\log\log k$

Wiki. Art01, §4, writes

$$\sum_{i\le k}\log\log p_i \ge k\left(\log\log k + \frac{\log\log -3/2}{\log k}\right) \qquad (k \ge 6),$$

with a bare $\log\log$ in the numerator.

Source. De Koninck–Letendre, arXiv:1812.09950, Lemma 4.8 (4.21):

$$\sum_{i\le k}\log\log p_i \ge k\left(\log\log k + \frac{\log\log k - 3/2}{\log k}\right) \qquad (k \ge 6).$$

Proposed correction. Replace the bare $\log\log$ by $\log\log k$.

1.5 Büthe 2016, omitted π^* bound

Wiki. Art01, §1, under Büthe (2016), lists RH-conditional bounds for ψ, ϑ, and π only.

Source. Büthe also states

$$\bigl|\pi^*(x) - \mathrm{li}(x)\bigr| \le \tfrac{\sqrt{x}}{8\pi}\log x \qquad (x > 59),$$

under the same height hypothesis.

Proposed correction. Add the π^* bound to the displayed list.

1.6 Axler 2019, Mandl B_n mislabelled as $\sum p_i$

Wiki. Art01, §4, attributes to Axler (2019) the bounds

$$\sum_{i\le k} p_i \gtrless \frac{k^2}{4} + \frac{k^2}{4\log k} - \frac{k^2(\log\log k - c)}{4(\log k)^2}$$

(with $c = 2.9$ for the lower bound at $k \ge 6309751$, and $c = 4.42$ for the upper bound at $k \ge 256376$).

Source. Axler, *On the sum of the first n prime numbers*, arXiv:1606.06874, Theorems 1.6 and 1.7 bound Mandl's quantity

$$B_n = \frac{n\,p_n}{2} - \sum_{k\le n} p_k,$$

not $\sum p_k$. The actual prime-sum estimates are Theorems 1.8 and 1.9 (leading factor $n^2/(2\log n)$, different constants and ranges).

Proposed correction. Replace $\sum_{i\le k} p_i$ by B_k (and cite Theorems 1.6–1.7), or else quote Theorems 1.8–1.9 for the prime-sum bounds.

1.7 Cosmetic / typesetting (lower priority)

- Art01 opening paragraph: "van Mangold" → "von Mangoldt"; "direction verifications" → "direct verifications".
- Art01, Ramaré (2013) second display: unmatched parenthesis in the O^* term.
- Art01, Cipolla expansion: mixes ln and log.
- Art01, Kadiri (2008) table (§6): the entry under $\epsilon = 1$, $q_0 = 10^{25}$ is rendered as ".5116"; almost certainly "4.5116".

2 Art09 — Short intervals containing primes

2.1 Carneiro–Milinovich–Soundararajan 2019, interval orientation

Wiki. Art09 quotes, under RH and for $x \ge 4$, a prime in

$$\big]x - \tfrac{22}{25}\sqrt{x}\log x,\; x\big].$$

Source. Carneiro–Milinovich–Soundararajan, *Fourier optimization and prime gaps*, Comment. Math. Helv. 94 (2019); arXiv:1708.04122, Theorem 5:

$$\big[x,\; x + \tfrac{22}{25}\sqrt{x}\log x\big].$$

(The backward interval of that shape is Dudek's earlier result with constant $4/\pi$; Carneiro et al. improve the constant and reverse the orientation.)

Proposed correction. Replace the displayed interval by $[x,\, x + \frac{22}{25}\sqrt{x}\log x]$.

2.2 Trudgian 2016, Corollary 2 threshold

Wiki. Art09 quotes Trudgian (2016) for $x >$ 2,898,242.

Source. Trudgian, *Updating the error term in the prime number theorem*, Ramanujan J. 39 (2016); arXiv:1401.2689, Corollary 2: $x \ge$ 2,898,239.

Proposed correction. Replace 2,898,242 by 2,898,239, and use the non-strict threshold $x \ge \cdots$ as in the paper.

2.3 Dudek 2014/2016, short-interval threshold

Wiki. Art09 quotes Dudek for a prime in $(x, x + 3x^{2/3}]$ when $x \ge \exp(\exp(34.32))$.

Source. Dudek, *An explicit result for primes between cubes*, Funct. Approx. Comment. Math. 55 (2016); arXiv:1401.4233, proves the short-interval claim under

$$x^{1/3} > \exp(\exp(33.217))$$

(equivalently $x > \exp(3\exp(33.217))$), as the x-form of the main cubes theorem $n \geq \exp(\exp(33.217))$. The wiki value 34.32 is a rounded stand-in for $\ln 3 + 33.217 \approx 34.3156$ and is slightly too large.

Proposed correction. Replace the hypothesis by $x^{1/3} > \exp(\exp(33.217))$ (or the exact equivalent $x > \exp(3\exp(33.217))$).

2.4 Cully–Hugill 2021, short-interval threshold

Wiki. Art09 quotes Cully–Hugill for $(x, x + 3x^{2/3}]$ when $x \geq \exp(\exp(33.99))$.

Source. Cully–Hugill, *Primes between consecutive powers*, arXiv:2107.14468, states $x \geq \exp(\exp(33.990))$ for that interval (with $m = 3$).

Proposed correction. Replace 33.99 by 33.990.

3 Art01 — Ramaré 2013 (additional)

3.1 Ramaré, original corollary versus 2023 corrigendum

Wiki. Art01, after noting that "there was an error in this paper which is corrected below", quotes for $x \geq 1\,520\,000$ the error $O^*(0.0067/\log x)$, and for $x \geq 468\,000$ the error $O^*(0.01/\log x)$.

Source. The original Ramaré corollary (Acta Arith. 159 (2013)) stated $O^*(0.0067/\log x)$ for $x \geq 23$, with no $0.01/\log x$ line. The 2023 corrigendum (sign errors and a computer-program typo for checks with $x \leq 10\,000$) restates the corollary with $x \geq 1.52 \cdot 10^6$ for $0.0067/\log x$, $x \geq 468\,000$ for $0.01/\log x$, and an additional line $O^*(1/(4\log x))$ for $x \geq 115$. The wiki matches the corrigendum (but omits the $x \geq 115$ line).

Proposed correction. Cite the 2023 corrigendum next to the revised thresholds, mention that the 2013 printed corollary had $x \geq 23$ for $0.0067/\log x$, and add the $x \geq 115$ bound.

4 Art06 — Bounds on $|\zeta(s)|$

4.1 Patel 2022, first term $(3/4)\log t$ versus $\log t$

Wiki. Art06 quotes

$$|\zeta(1+it)| \leq \min\Big(\tfrac{3}{4}\log t,\ \tfrac{1}{2}\log t + 1.93,\ \tfrac{1}{5}\log t + 44.02\Big) \qquad (t \geq 3).$$

Source. Patel, arXiv:2009.00769, Theorem 1.1:

$$|\zeta(1+it)| \leq \min\big(\log t,\ \tfrac{1}{2}\log t + 1.93,\ \tfrac{1}{5}\log t + 44.02\big).$$

The factor $\frac{3}{4}$ belongs to Trudgian's earlier bound, also listed on Art06.

Proposed correction. Replace $\frac{3}{4}\log t$ by $\log t$ in the Patel display.

4.2 Hiary 2016 constant 0.63

Wiki. Art06 quotes $|\zeta(1/2+it)| \le 0.63\, t^{1/6} \log t$ for $t \ge 3$, citing Hiary (2016).

Source. That constant used an incorrect form of the Kusmin–Landau lemma. Hiary–Patel–Yang, arXiv:2207.02366, note that the corrected Hiary constant is 0.77, and prove the improved bound with constant **0.618**.

Proposed correction. Replace 0.63 by 0.618 and cite Hiary–Patel–Yang, or else state explicitly the corrected Hiary constant 0.77.

4.3 Rosser 1941, threshold $T \ge 2$ versus $T \ge 1467$

Wiki. Art06 quotes Rosser's $N(T)$ bound for $T \ge 2$.

Source. Rosser (1941), as recorded e.g. in Trudgian's tables, proves $|S(T)| \le 0.137 \log T + 0.443 \log\log T + 1.588$ for $\mathbf{T \ge 1467}$.

Proposed correction. Replace $T \ge 2$ by $T \ge 1467$.

4.4 Trudgian 2014, extraneous $1/(5T)$

Wiki. Art06 appends $+1/(5T)$ to the $N(T)$ error for Trudgian (2014).

Source. The published result (J. Number Theory 134) is $|S(T)| \le 0.112 \log T + 0.278 \log\log T + 2.510$ for $T \ge e$, with no $1/(5T)$ term. (An arXiv corollary has a constant $0.2/T_0$ when $T \ge T_0 \ge e$, which is not $1/(5T)$.)

Proposed correction. Drop the $1/(5T)$ summand (or replace it by the parameterized $0.2/T_0$ form, citing the corollary carefully).

4.5 Lehman 1970, $t \ge 1/5$ versus $64/(2\pi)$

Wiki. Art06 states Lehman's bound for $t \ge 1/5$, noting that Lehman wrote $t \ge 64/(2\pi)$ and that $1/5$ is a modern computational extension.

Source. Lehman, Lemma 2: $t \ge 64/(2\pi)$.

Proposed correction. Quote $t \ge 64/(2\pi)$ for the paper statement; keep $t \ge 1/5$ only if explicitly labelled as a verified extension.

4.6 Delange 1987, complex versus real

Wiki. Art06 quotes, for $\sigma > 1$ and any real t,

$$-\Re \frac{\zeta'}{\zeta}(\sigma + it) \le \frac{1}{\sigma - 1} - \frac{1}{2\sigma^2}.$$

Source. Delange, Colloq. Math. 53 (1987), proves for **real** $s > 1$

$$\frac{\zeta'}{\zeta}(s) + \frac{1}{s-1} > \frac{1}{2s^2}$$

(equivalently the displayed upper bound on $-\zeta'/\zeta(s)$ on the real line). The complex form is not in that note.

Proposed correction. Restrict the statement to real $s > 1$, or cite a different source for the complex claim.

4.7 Hasanalizade–Shen–Wong $N(T)$ forms

Wiki / transcription risk. Art06 lists HSW parameters 0.1038, 0.2573, 9.3675 in the shape that includes the classical 7/8 term.

Source. Hasanalizade–Shen–Wong, arXiv:2107.06506:

- Corollary 1.2 (no 7/8): constant term 9.3675, for $T \geq e$;
- Corollary 1.4 (bound on $S(T) = N(T) - \text{main} - 7/8$): constant terms 8.3675 and 3.0305.

Pairing 9.3675 with the +7/8 form conflates the two corollaries.

Proposed correction. Display Cor. 1.2 and Cor. 1.4 separately, with the matching main terms.

5 Art08 — Zero-free regions and related results

5.1 Classical zero-free region written with $\log(|\Im s| + 2)$

Wiki. Art08 quotes Kadiri (2005), Mossinghoff–Trudgian (2015), and the later refinement with $R_0 = 5.558691$, all in the shape

$$\Re s \geq 1 - \frac{1}{R_0 \log(|\Im s| + 2)}.$$

Source. The cited papers use $\log |\Im s|$ (equivalently $\log |t|$), not $\log(|t| + 2)$:

- Kadiri, *Une région explicite sans zéro pour la fonction Zeta de Riemann*, arXiv:math/0401238: main theorem $\Re s \geq 1 - 1/(5.70176 \log |\Im s|)$ for $|\Im s| \geq 2$.
- Mossinghoff–Trudgian, arXiv:1410.3926, Theorem 1: $\sigma \geq 1 - 1/(5.573412 \log |t|)$ for $|t| \geq 2$.
- Mossinghoff–Trudgian–Yang, arXiv:2212.06867: $\sigma \geq 1 - 1/(5.558691 \log |t|)$ for $|t| \geq 2$.

(The same $\log |t|$ form is the one used for the classical zero-free region predicate in the Lean formalization, following Fiori–Kadiri–Swaminathan.)

Proposed correction. Replace $\log(|\Im s| + 2)$ by $\log |\Im s|$ in each of these displays (keeping the stated ranges $|t| \geq 2$).

5.2 Kadiri R_0: 5.70175 versus 5.70176

Wiki. Art08 attributes to Kadiri (2002/2005) the constant $R_0 = 5.70175$.

Source. Kadiri, arXiv:math/0401238, states the main theorem with $R_0 = \mathbf{5.70176}$. (The digit 5.70175 appears elsewhere in numerical tables in that paper, but is not the constant of the theorem statement quoted on Art08.)

Proposed correction. Replace 5.70175 by 5.70176.

Status in the Lean formalization

The corresponding statements in `PrimeNumberTheoremAnd/IEANTN/TMEEMT.lean` and `SecondarySummary.lean` have been aligned with the source papers for items 1.1, 1.2, 1.3, 1.4, 1.6, 2.1, 2.2, 2.3, and 2.4. Item 1.1 also required restoring the factor 2 in the π_2 main term $\frac{2x^3}{27\log^3 x}$, which Art01 already had correctly but which had been dropped in an earlier transcription into Lean. Item 1.6 was likewise present as a Lean transcription error (copied from the wiki); the formal statements now bound Mandl's B_n as in Theorems 1.6–1.7 of Axler. Items 2.1, 2.2, 2.3, and 2.4 were present both on Art09 and in Lean; all now follow the papers. Items 5.1 and 5.2 are wiki-only: the project's `riemannZeta.classicalZeroFree` already uses $\log|\Im s|$ without the spurious $+2$, matching the papers and Fiori–Kadiri–Swaminathan. Items 4.2, 4.7, 4.1, 4.3, 4.4, 4.5, and 4.6 are aligned in Lean in a companion pull request. Item 3.1: Lean follows the 2023 corrigendum thresholds (including the $x \geq 115$ line omitted on the wiki).